ANSHUL PARATE

AI Developer | Full-Stack & Machine Learning Engineer
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Professional Summary

AI Developer and Full-Stack Machine Learning Engineer with experience building production-grade LLM applications, multi-modal RAG pipelines, and agentic workflows. Proven track record in developing scalable backends with FastAPI and PostgreSQL, designing responsive React.js interfaces, and deploying high-accuracy Computer Vision and NLP models. Strong algorithmic problem-solving background focused on low-latency inference, clean architecture, and delivering high-impact engineering solutions.

Education

Shri Ramdeobaba College of Engineering & Management, Nagpur *2023 – Present*
B.Tech CSE (AIML), CGPA: 8.05 / 10

Prerna International School, Vihirgaon, Nagpur *2022*
HSC (77.6%)

Sanskar Vidya Sagar, Nagpur *2020*
SSC (80.8%)

Technical Skills

- **Generative AI & Agents:** LangGraph, LangChain, OpenAI Agent SDK, MCP Server, Multi-Modal RAG, Vector Stores
- **Machine Learning & CV:** Computer Vision, OpenCV, YOLOv8, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **Backend Development:** Python, FastAPI, RESTful APIs, Node.js, Microservices, Asynchronous Programming
- **Databases & Stores:** PostgreSQL, MongoDB, Qdrant, Chroma, Neo4j, Data Modeling, Query Optimization
- **Frontend Development:** React.js, TailwindCSS, Responsive Design, State Management, UI/UX Design, Figma
- **Cloud & DevOps:** Git, GitHub, Docker, Vercel, Render, Linux, Postman, CI/CD Basics

Projects

MVL-AgenticAI (Vectorless Hierarchical Multimodal RAG)

- Architected a hierarchical vectorless agentic RAG system with LangGraph, reducing indexing tokens by 76%–90% (38K vs. 161K) and cutting API calls by 95% (4.7 vs. 98.3) over baselines.
- Engineered cross-modal fusion binding high-res figures, HTML tables, and audio streams into a unified tree, improving retrieval precision (MRR@5) from 0.65 to 0.90 with zero vector DB cost.
- Developed an autonomous tree-navigation retriever with dynamic fallback mechanisms, achieving 0.814 structural hierarchy fidelity across complex multi-page filings.

Tech: LangGraph, Multi-Agent Architecture, Python, FastAPI, React.js, Multimodal Vision | **Links:** Website | GitShare

Renewal Upsell Advisor Platform

- Designed a full-stack recommendation engine analyzing 10,000+ customer data points for high-probability upsells.
- Orchestrated automated workflows and REST APIs, improving conversions by 35% and cutting review time by 40%.

Tech: Python, React.js, FastAPI, LangGraph, PostgreSQL | **Links:** GitHub | Website

Financial Adviser (NLP Strategy & Backtesting Engine)

- Architected an NLP strategy generator parsing natural language into an executable DSL → AST pipeline with 95% accuracy.
- Integrated predictive ML models alongside historical backtesting over 5+ years of market data to forecast signals with 88% accuracy.

Tech: React.js, TailwindCSS, FastAPI, NLP, Python, Predictive ML | **Links:** GitHub | Website

Experience

AI Lifebot | *AI / Full-Stack Intern* *Feb 2026 – Jul 2026*

- Delivered 2+ production full-stack AI applications, managing end-to-end LLM integration, prompt engineering, and React dashboards.
- Collaborated with a 5-member cross-functional engineering team in 2-week Agile sprints with code reviews and CI/CD.
- Diagnosed bottlenecks and optimized API response times, ensuring seamless communication between client and backend.

Atomrank.in | *AI Engineering Intern* *Aug 2025 – Feb 2026*

- Built high-throughput backend services and REST APIs, scaling to ~15,000 requests/min with PostgreSQL logging.
- Integrated OpenAI Agent SDK over 6+ NCERT textbooks with Qdrant vector database for low-latency contextual search.
- Wrote clean, modular code following test-driven practices, structured debugging, and git-based version control.

Achievements

- **Winner — Hackathon '26 (TECHBLONS 2026):** Won 1st prize for "Evolution of AI/ML in Education", organized by P. R. Pote Patil College of Engg. & Mgmt., Amravati – **Certificate**.
- **SIH College-Round Qualifier:** Qualified in National Smart India Hackathon for open-pit mine risk detection.